# SnapPay Platform Transformation

## Enhancing Scalability, Performance, Cost Reduction & One Platform

Ranganathan, SK

# Agenda

- Overview of SnapPay and reasons for modernization
- Future vision for SnapPay
- Transformation strategy
- Current architecture and challenges (47 DB calls per charge, TPS 17)
- Q&A session
- Proposed architecture with increased TPS (1000) and cost savings
  - Horizon 1: Foundation Setup (June 2026)
  - Horizon 2: Web Platform Modernization (Aug 2026)
  - Horizon 3: API modernization (March 2027)
  - Horizon 4: Batch jobs, Dashboards and Data migration (June 2027)


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# Overview of SnapPay and reasons for modernization

## SnapPay Today

- Mission-critical enterprise payments platform supporting cards, ACH, invoicing, recurring payments, and customer self-service
- Serves multiple segments (Enterprise, eCommerce, Government) across two production platforms
- Built as a large, tightly coupled monolithic application on a legacy technology stack
- High operational risk, slow change velocity, and limited scalability

## Why Modernization Is Needed

- Current architecture limits scalability, resilience, and speed to market
- Fragile data and deployment model increases operational risk
- Difficult to evolve UI, APIs, integrations, and onboarding flows independently
- Growing business demand requires a more flexible, cloud-ready platform


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# Future Vision for SnapPay

Target Vision

- Modular, cloud-ready platform with clear service and data boundaries
- API-first design enabling modern UI, partner integrations, and extensibility
- Active/active architecture with near-zero downtime deployments
- Strong security, observability, and operational automation built in
- Ability to onboard merchants faster and migrate existing merchants safely

Non-Negotiables

- No customer-facing URL changes
- No breaking API changes
- Zero disruption to existing integrators
- Parallel run and safe cohort-based migration
- Zero downtime during deployments.


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# Transformation Strategy

## Modernization Approach

- Incremental, phased modernization — no big-bang rewrite
- MVP first: production-ready platform that can onboard and operate merchants
- Parallel run with controlled migration and rollback
- Ongoing feature development continues throughout modernization

## Key Principles

- Vendor-proposed architecture within clear guardrails
- Backward compatibility as a release gate
- Automation-first: CI/CD, testing, security scanning, observability
- Security and operational readiness are mandatory, not optional
- 99.995% availability,
- 500 TPS sustained, 1000 TPS peak, <2s API latency
- Safe migrations with proven rollback
- Faster delivery with lower operational risk

## Accelerators & AI-Assisted Engineering

- Leverage AI-assisted accelerators for unit test generation, refactoring, and code modernization to improve speed and quality.
- Enforce automated code review and quality checks within CI/CD, including security, performance, and design-pattern validation
- Implement code coverage measurement and gating for new and modernized components
- Use reusable templates, scaffolding, and reference implementations aligned with the target architecture to ensure consistency and productivity.

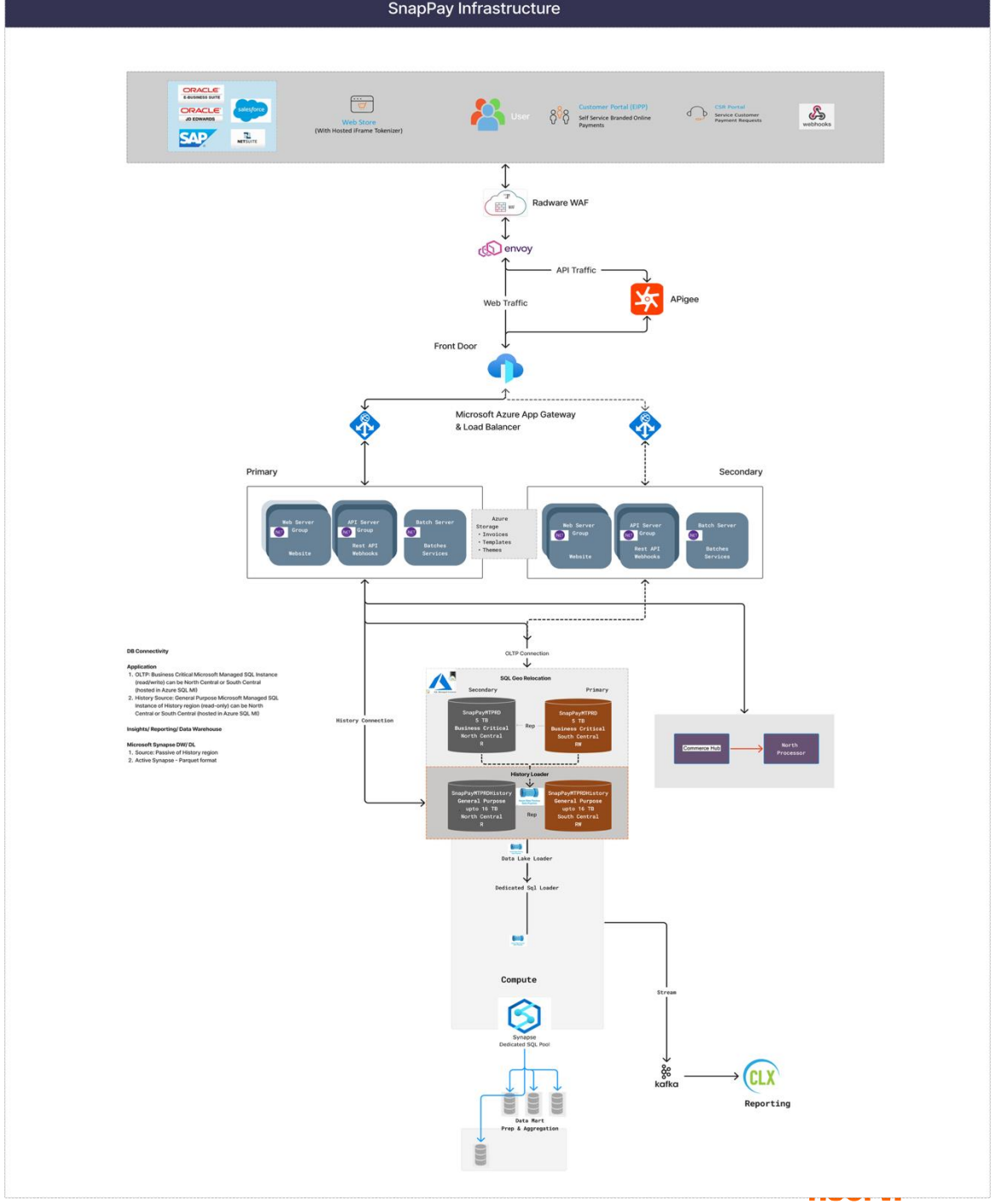

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# SnapPay Current Infrastructure & Architecture


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# Infrastructure Diagram



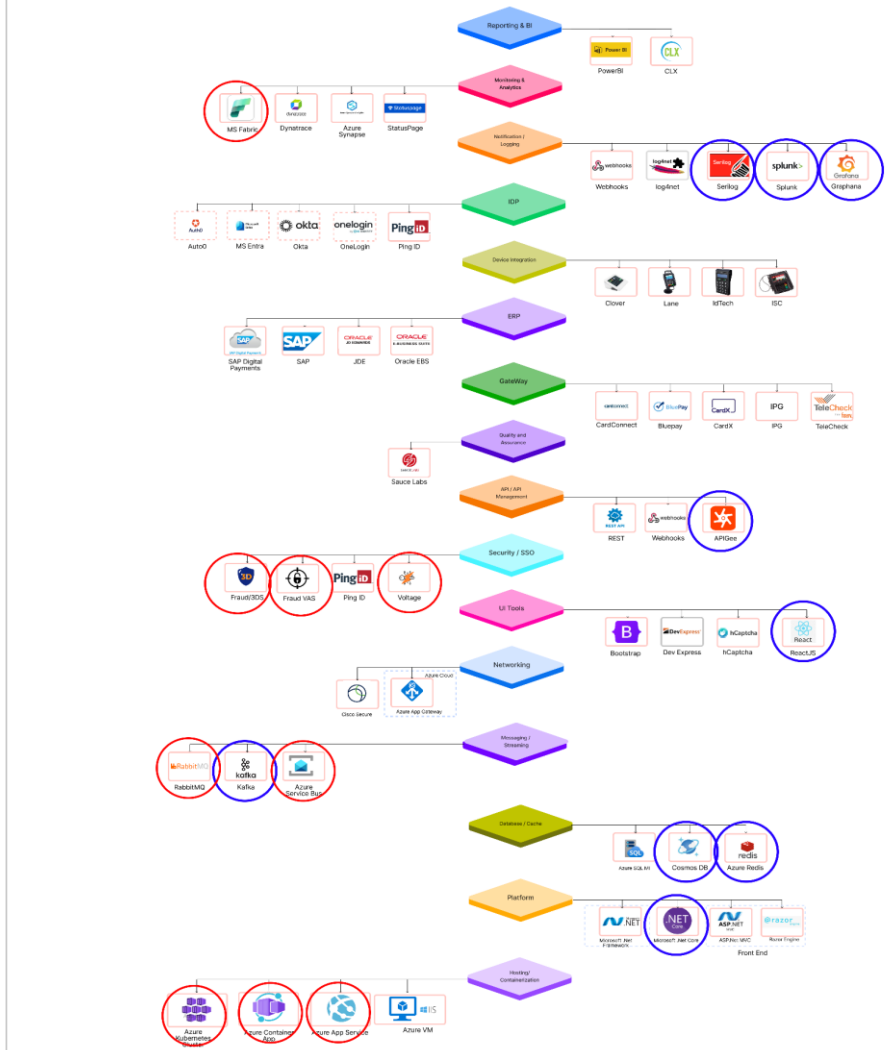


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# Technology Stack

## SnapPay Technology Stack

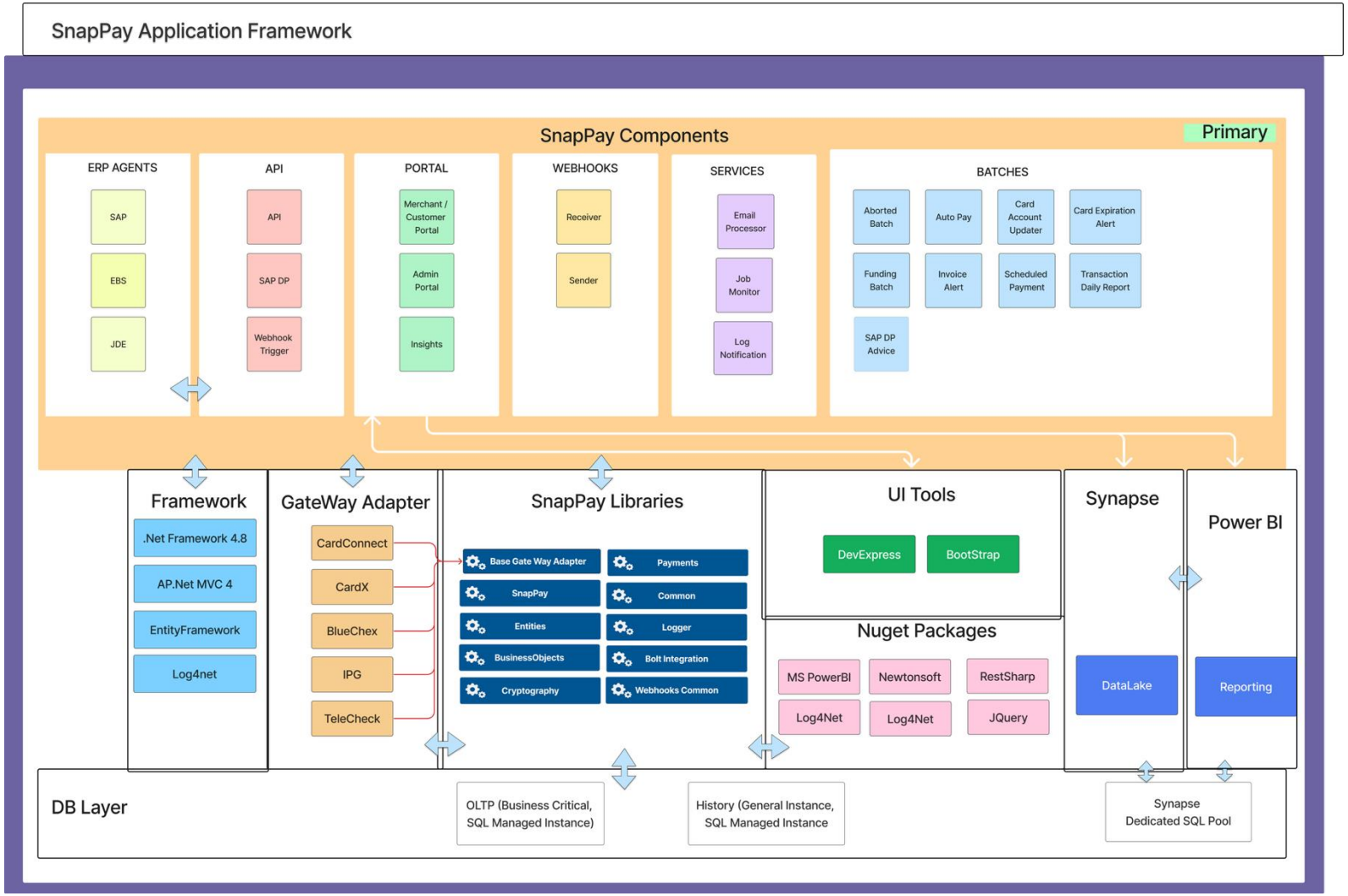




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# Application Architecture Diagram

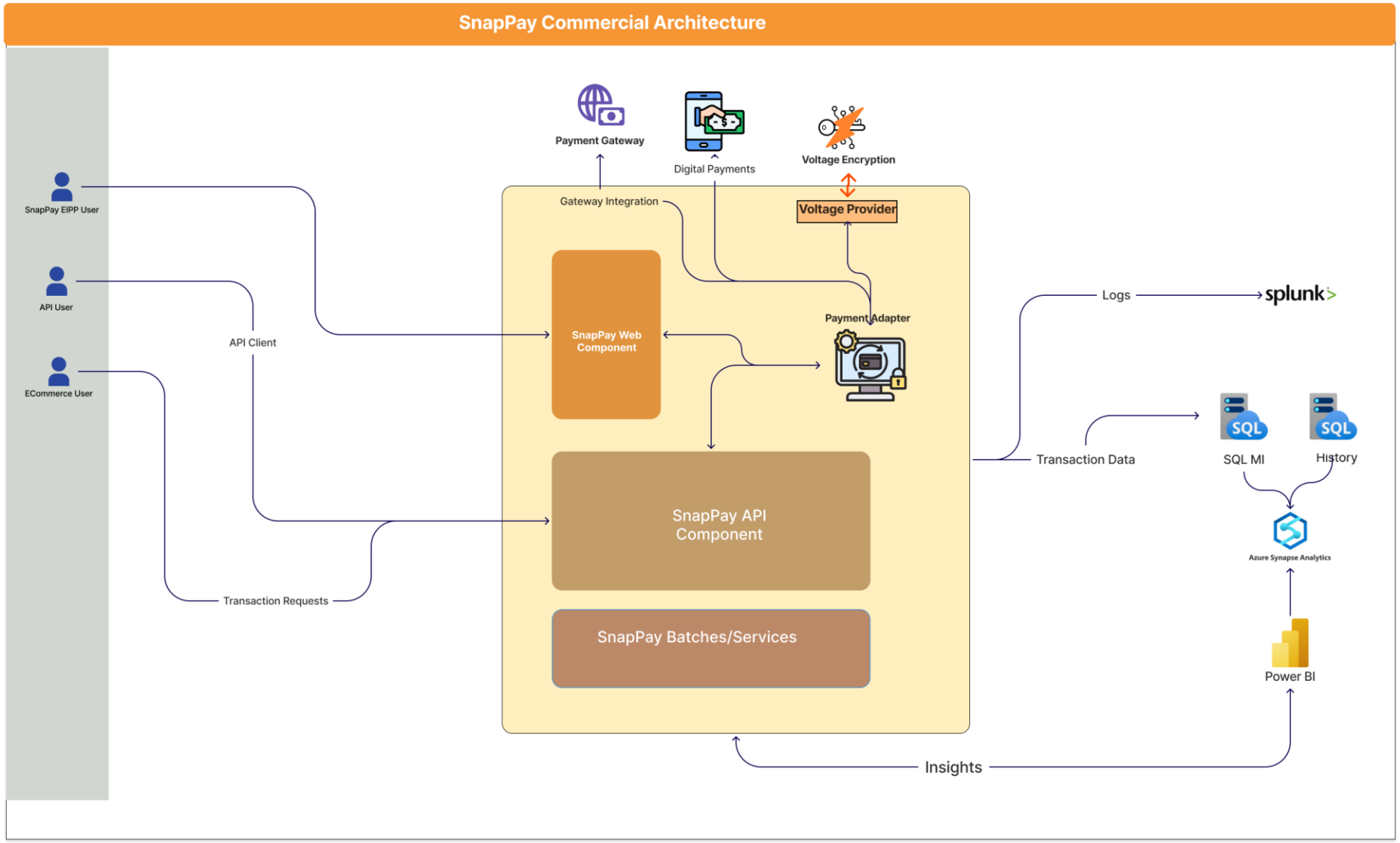




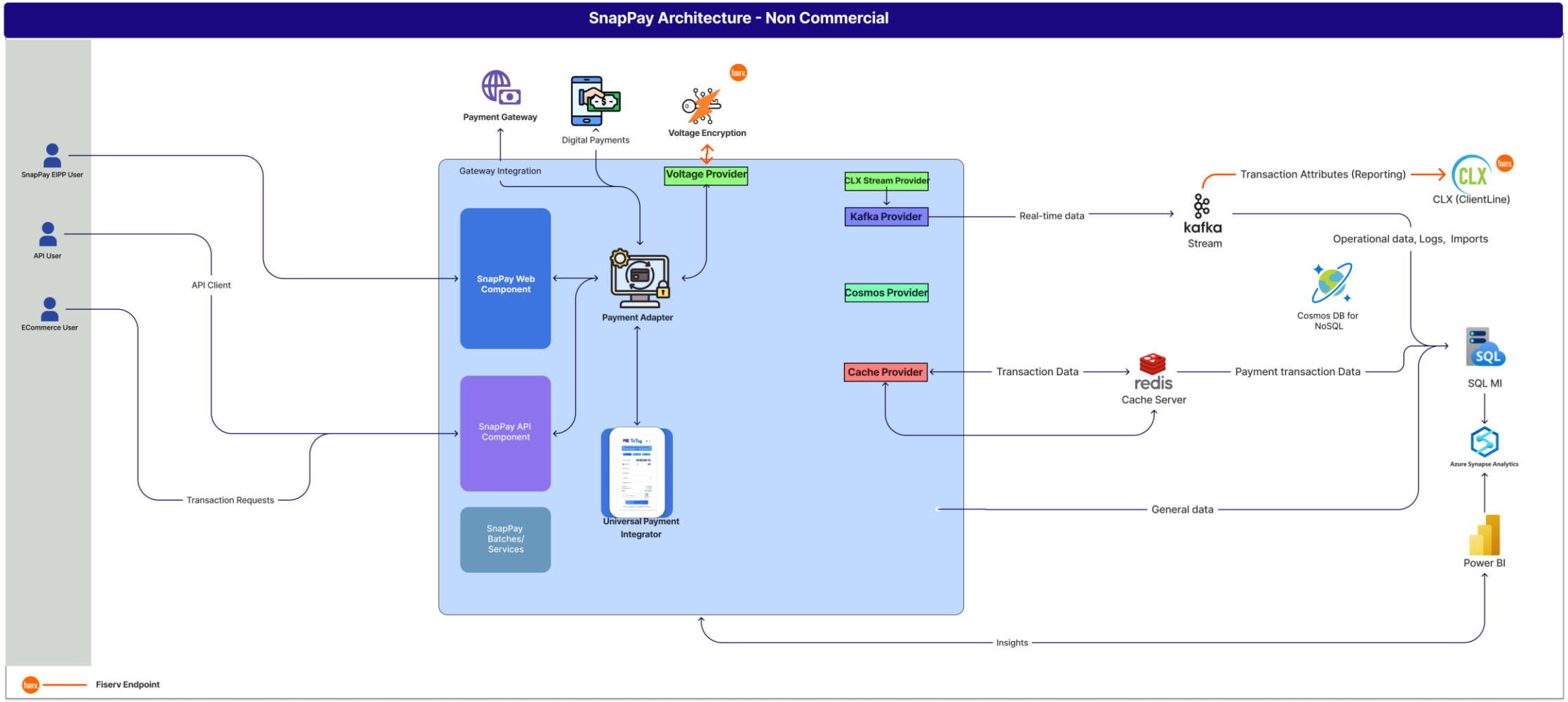
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# Existing Application Architecture (Commercial Landscape)



# Existing Application Architecture (Non Commercial Landscape)





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# Current Challenges, Pain Points

- Monolithic Coupling & Deployment Bottlenecks
  - 53 projects in a single solution - Any change requires full solution rebuild/deployment (~30 min)
  - Tightly coupled payment providers (CardConnect, CardX, BluePay, TeleCheck) - Can't deploy or scale providers independently
  - Single point of failure - One provider bug can crash entire payment processing system
- Performance Bottlenecks & Database Inefficiencies
  - 17 TPS
  - No caching layer - Every payment lookup hits database directly (no Redis/distributed cache)
  - High number of DB Calls (e.g Charge API has around ~40+ DB calls)
  - Cross-database validation, increase latency, prevent database-per-service pattern
- Scalability & Cost Constraints
  - SQL Managed Instance - ~$20K/month minimum, scales vertically only (instance restart required)
  - No horizontal scaling - Can't add more web servers without complex load balancer setup
  - Total Infrastructure Cost: ~$1.2M/year (vs $800K with Modernization = ~20% savings)
  - Non-Comm – $40K per month, Comm- $45K per month
- Obsolete Technology Stack & Technical Debt
  - DOTNET Framework 4.7.2 - End of Support October 2027 (18 months away), no new features
  - EF6 EDMX (Database First) - Manual EDMX regeneration for every schema change (error-prone, 15 min process)
- Maintenance Burden & Operational Complexity
  - No automated migrations - Schema changes require manual SQL scripts + EDMX regeneration + code updates
  - Hard to onboard developers - New team member needs 3 weeks to understand payment flow
- Limited Observability & Debugging Challenges
  - No distributed tracing - Can't track payment transaction across API → Provider → Gateway → Database
  - No centralized monitoring - Each batch process logs to local file, requires RDP to troubleshoot


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# High Level Approach & Timeline

## Horizon 1 Foundation Setup

Initial phase includes database migration, architectural approval, and building the APIs needed for Horizon 2 (App modernization).

April – June 2026

## Horizon 2 Web Portal Modernization

Horizon 2 focuses on deploying the modernized EIPP system meeting functional, performance, and security standards.

June – Aug 2026

## Horizon3 API, Batch Jobs Modernization

Horizon 3 upgrades SnapPay V1-V3 APIs to microservices, ensuring consistent, modular, and scalable production operations.

Aug – March 2027

## Horizon4 Dashboards & Migration

Final phase ensures safe merchant migration using cohort-based methods, reconciliation gates, and rollback protections.

March – June 2027


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# Elements of Current SnapPay

| Component Name | Count |
|---|---|
| REST API Endpoints | 60-70 |
| Batch Process | 10 |
| Windows Services | 3 |
| Web Screens* | 20-30 |
| Payment Providers | 6 |
| Database Objects | 350-400 |
| Webhook Endpoints | 3 |
| Test Cases (regressions, automated, etc.,) | 750 |

* - Done by Fiserv FTE


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# Horizon –1, 2

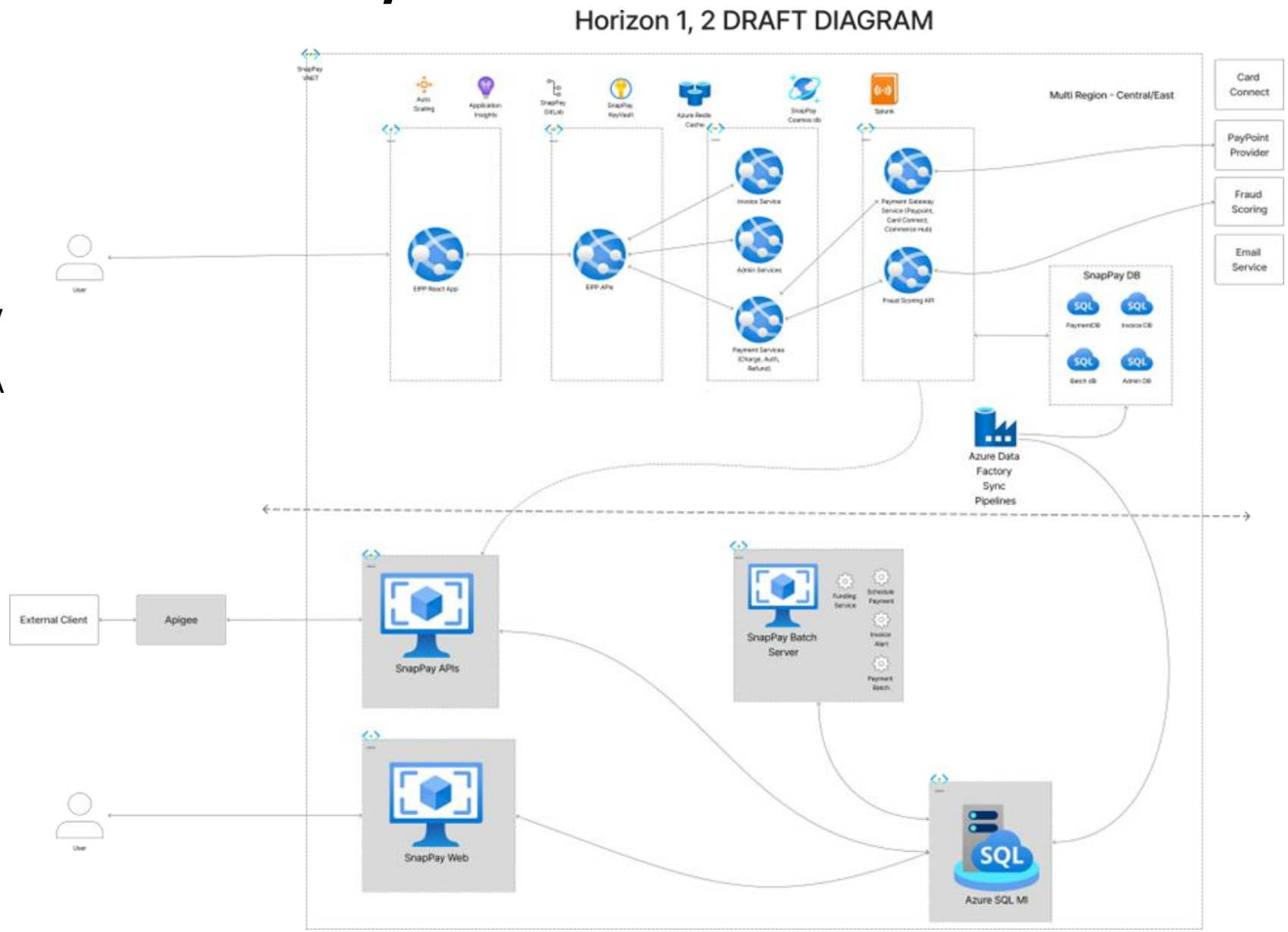
# Foundation & SnapPay EIPP Modernization To Azure with Cloud Native Services

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# Horizon 1,2 – Move to Azure PaaS, EIPP Modernization

- Stepwise Modernization, IaaS > PaaS> CaaS, Foundation for Containerization
- Design & Implement stateless microservices on Linux containers, multi-region(central/East)
- Use Managed Identity to connect to Azure resources
- All secret/certificates should be stored in Key Vault
- Distributed tracing, resilience pattern with HA and Autoscaling
- Split Monolithic database into multiple database
- Get all ARB(Architecture Review Board) approval, follow Fiserv standards
- Setup Infra, CI/CD pipeline, automation
- Complete QA, Performance Testing, UAT and delivered to customer on production environment
- EIPP App hosted on azure cloud native services and meeting all SnapPay Transformation principles




Gray color boxes - existing components


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# Horizon -3

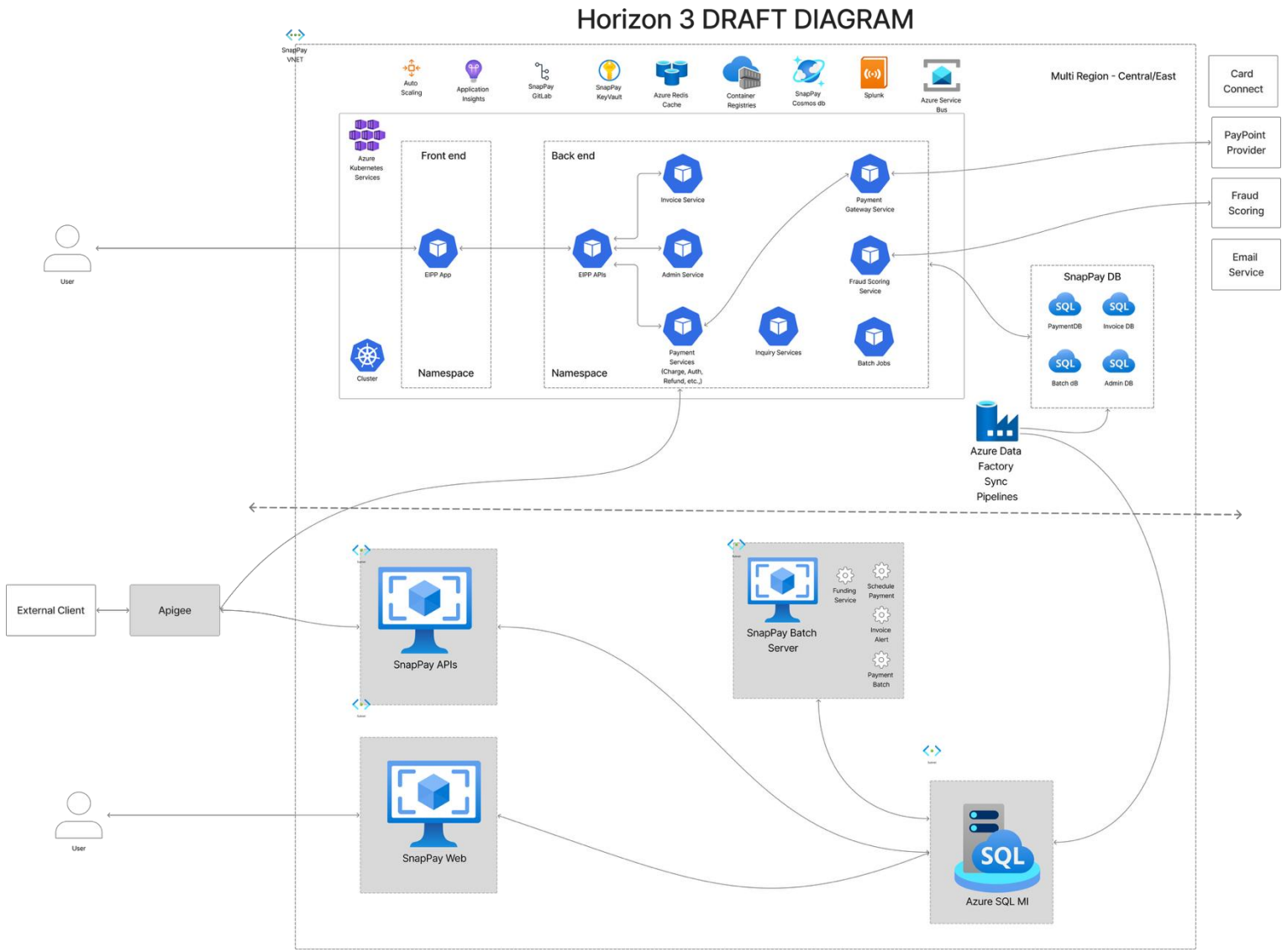
# Move to Containerization,  Remediation of SnapPay Components to Microservice Architecture


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# Horizon 3– Move to Containerization, Remediation of SnapPay Components to Microservice Architecture

- Move from PaaS to CaaS
- Utilize Fiserv AKS for Container as a Service
- Remediate existing SnapPay REST APIs based on microservice framework setup in Horizon 1,2
- Move to Event Based Architecture using Azure Service Bus
- Batch jobs, services, webhooks, etc moved to new architecture
- Connectivity to SnapPay Core Micro Services, EIPP Web Portal is set up through the FISERV Multi-Cloud Backbone, providing robust and reliable inter-service communication
- SnapPay Core Micro Services, Web Portal, and other services are containerized and deployed in Fiserv Azure Kubernetes Cluster. These microservices are distributed across multiple regions (Central and East) for high availability and fault tolerance
- Legacy API dependencies reduced or eliminated
- Fully automated deployments on one single platform
- 500 TPS sustained, 1000 TPS peak, <2s API latency

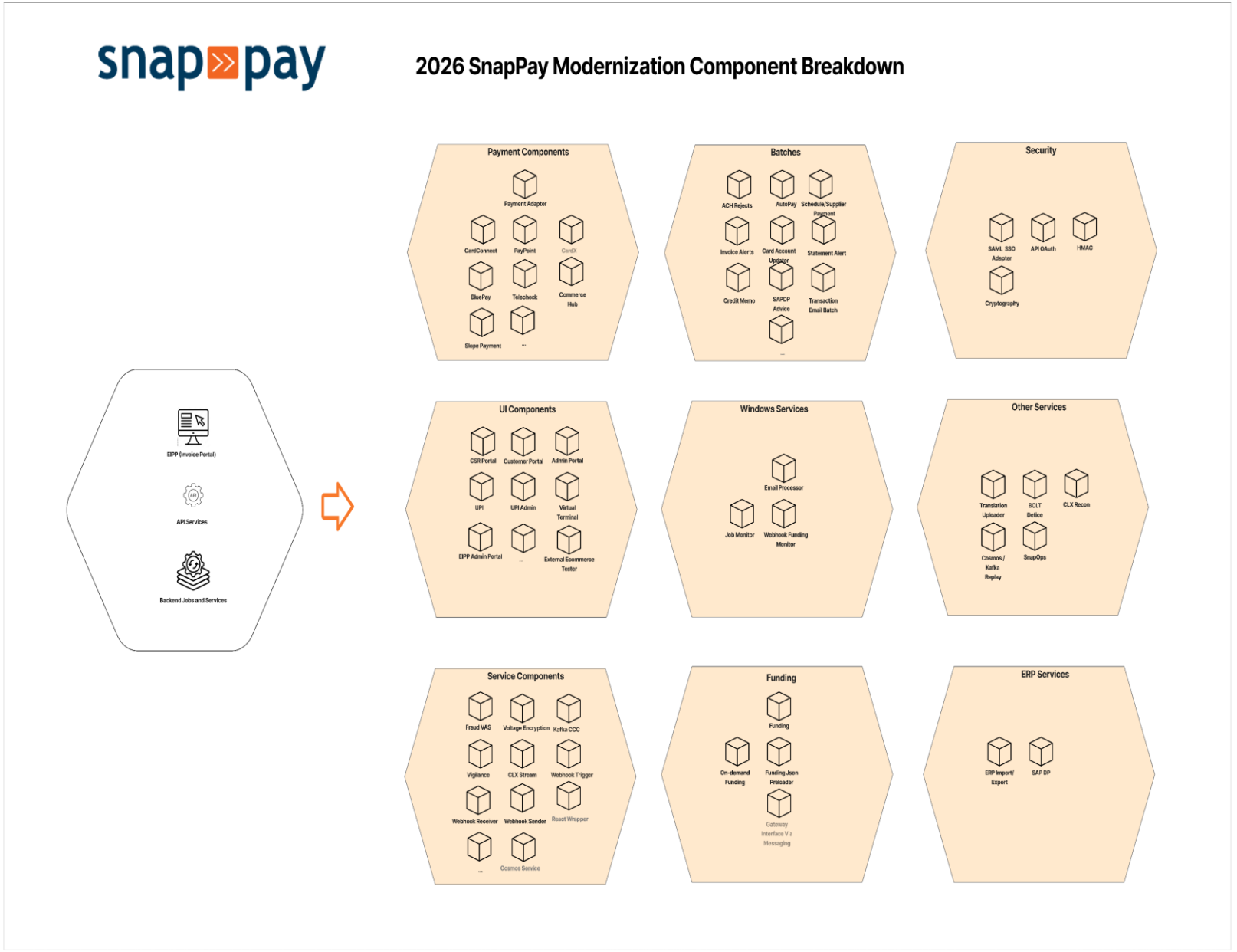



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# SnapPay Modernization Component Breakdown




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# Horizon -4

# Migrating existing customers, one unified platform

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# Horizon 4– Details

TBD

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# SnapPay Platform Transformation Benefits

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# Business & Technology Benefits

- Lower platform costs (20–40%) through modular, cloud-ready architecture, elastic scaling, and platform consolidation
- Higher availability & resilience (99.995%) with near-zero downtime deployments and active/active design
- Faster delivery (25–40%) via API-first, modular services and CI/CD automation

- Reduced operational risk with automated testing, observability, and backward-compatibility release gates
- Improved security posture through DevSecOps, automated vulnerability scanning, and stronger isolation
- Faster merchant onboarding (30–50%) enabling earlier time-to-revenue
- Avoided migration risk through incremental modernization, parallel run, and proven rollback

Outcome: A safer, scalable, and cost-efficient SnapPay platform that supports growth without customer disruption.


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# Post Go-Live KPIs to Track Transformation Benefits

## Cost & Efficiency

- Infrastructure cost per transaction
- Total platform run cost (baseline vs. post-modernization)
- % idle vs. utilized capacity
- Cost savings from legacy platform decommissioning

## Reliability & Operations

- Availability (% uptime)
- Mean Time to Detect (MTTD)
- Mean Time to Recover (MTTR)
- Number of Sev-1 / Sev-2 incidents per quarter
- Deployment failure rate / rollback frequency

## Delivery & Productivity

- Release frequency (per biweekly/month)
- Lead time for change (commit → production)
- Change failure rate
- Regression defects per release

## Security & Compliance

- Critical/high vulnerabilities at release (target: zero)
- Time to remediate vulnerabilities
- Security findings per release

## Migration & Adoption

- % merchants onboarded to new platform
- Migration success rate (no rollback)
- Time per merchant migration
- Customer-reported issues post-migration

## Growth Enablement

- Average merchant onboarding time
- New partner/API integrations per quarter
- Revenue realization time for new merchants


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# POD Structure

### SnapPay - Alpha

Edit

| | |
|---|---|
| Leads | 2 |
| Developers | 3 |
| QA Engineers | 1 |
| Velocity (pts/sprint) | 32 |
| Total Team | 6 |
| Annual Capacity | 768 pts |

▶Show Team Members 6

### SnapPay - Beta

Edit

| | |
|---|---|
| Leads | 1 |
| Developers | 3 |
| QA Engineers | 1 |
| Velocity (pts/sprint) | 30 |
| Total Team | 5 |
| Annual Capacity | 720 pts |

▶Show Team Members 6

### SnapPay - Titans

Edit

| | |
|---|---|
| Leads | 2 |
| Developers | 3 |
| QA Engineers | 2 |
| Velocity (pts/sprint) | 28 |
| Total Team | 7 |
| Annual Capacity | 672 pts |

▶Show Team Members 7

### SnapPay - Delta

Edit

| | |
|---|---|
| Leads | 1 |
| Developers | 0 |
| QA Engineers | 0 |
| Velocity (pts/sprint) | 30 |
| Total Team | 1 |
| Annual Capacity | 720 pts |

▶Show Team Members 3

### SnapPay - Admirals

Edit

| | |
|---|---|
| Leads | 3 |
| Developers | 1 |
| QA Engineers | 2 |
| Velocity (pts/sprint) | 34 |
| Total Team | 6 |
| Annual Capacity | 816 pts |

▶Show Team Members 6

### SnapPay - Zeta

Edit

| | |
|---|---|
| Leads | 2 |
| Developers | 3 |
| QA Engineers | 1 |
| Velocity (pts/sprint) | 30 |
| Total Team | 6 |
| Annual Capacity | 720 pts |

▶Show Team Members 8

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# Q&A

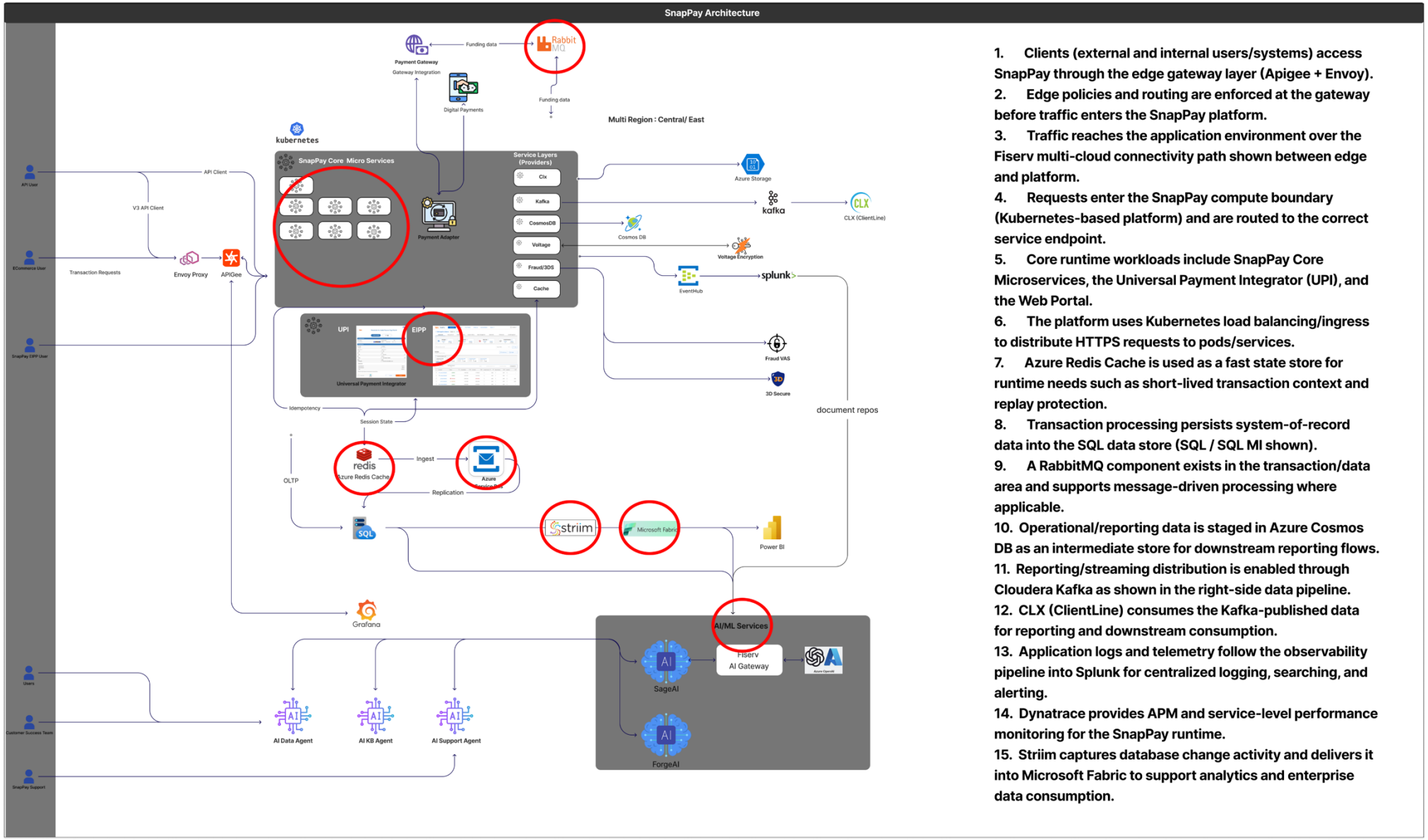
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# High-level Architecture




1. Clients (external and internal users/systems) access SnapPay through the edge gateway layer (Apigee + Envoy).
2. Edge policies and routing are enforced at the gateway before traffic enters the SnapPay platform.
3. Traffic reaches the application environment over the Fiserv multi-cloud connectivity path shown between edge and platform.
4. Requests enter the SnapPay compute boundary (Kubernetes-based platform) and are routed to the correct service endpoint.
5. Core runtime workloads include SnapPay Core Microservices, the Universal Payment Integrator (UPI), and the Web Portal.
6. The platform uses Kubernetes load balancing/ingress to distribute HTTPS requests to pods/services.
7. Azure Redis Cache is used as a fast state store for runtime needs such as short-lived transaction context and replay protection.
8. Transaction processing persists system-of-record data into the SQL data store (SQL / SQL MI shown).
9. A RabbitMQ component exists in the transaction/data area and supports message-driven processing where applicable.
10. Operational/reporting data is staged in Azure Cosmos DB as an intermediate store for downstream reporting flows.
11. Reporting/streaming distribution is enabled through Cloudera Kafka as shown in the right-side data pipeline.
12. CLX (ClientLine) consumes the Kafka-published data for reporting and downstream consumption.
13. Application logs and telemetry follow the observability pipeline into Splunk for centralized logging, searching, and alerting.
14. Dynatrace provides APM and service-level performance monitoring for the SnapPay runtime.
15. Striim captures database change activity and delivers it into Microsoft Fabric to support analytics and enterprise data consumption.

Proposed for 2026

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